Vedran Mitic

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PROFESSIONAL SUMMARY

Master's student in Artificial Intelligence and Machine Learning with a strong interest in quantitative analysis and financial data. Experienced in Python, SQL, and machine learning for extracting insights from complex datasets. Interested in applying data-driven methods to financial markets, portfolio analysis, and investment decision support.

EDUCATION

Faculty of Electronic Engineering, University of Nis — **Nis, Serbia**
Master's Degree in Artificial Intelligence and Machine Learning — *Nov 2025 – Present*

Faculty of Electronic Engineering, University of Nis — **Nis, Serbia**
Bachelor of Electrical Engineering and Computer Science (GPA: 8.22/10) — *Oct 2021 – Oct 2025*

Universidad Politécnica de Madrid — **Madrid, Spain**
Erasmus+ Student Mobility – ICT — *Feb 2025 – Jul 2025*

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, MATLAB
Big Data: Docker, Podman, Apache Spark, Hadoop (HDFS)
Machine Learning & Data: Supervised Learning, Unsupervised Learning, Data Preprocessing, Neural Networks, Scikit-learn, Pandas, NumPy, Explainable AI (XAI)
Tools: Git, Jupyter Notebook, PyCharm, VS Code

SELECTED PROJECTS

- **Strategic Transaction Segmentation & Profitability Analysis** — GitHub
  Developed an unsupervised ML pipeline to segment 9,000+ transactions and identify margin leakage patterns, translating raw data into actionable business insights.
- **Data Quality Engineering in Financial Predictive Modeling** — GitHub
  Analyzed the impact of Data Quality on ML model performance by designing a preprocessing and validation pipeline for bank customer churn prediction. Combined theoretical research with practical implementation in Python (Pandas, Scikit-learn, H2O Auto ML) to evaluate reliability and model stability.
- **Explainable Machine Learning on Biological Data** — GitHub
  Developed classification and regression models for biological datasets and applied SHAP and LIME to interpret model behavior and improve transparency.
- **ANN Architecture Optimization for Data Classification (Bachelor Thesis)** — GitHub
  Conducted empirical analysis of three-layer neural network architectures using MATLAB, comparing SCG and Levenberg–Marquardt optimization algorithms for binary and multi-class classification.
- **Big Data Infrastructure with Spark & Hadoop** — GitHub
  Designed and deployed a distributed data processing architecture using Docker, Hadoop (HDFS), and Apache Spark to process 4GB+ datasets, simulating scalable infrastructure for large-scale analytics workloads.

CERTIFICATIONS

- Red Hat OpenShift Development I: Introduction to Containers with Podman (DO188) – Red Hat, 2026

LANGUAGES

Serbian: Native **English:** C1 **Spanish:** A2 (Currently pursuing B1 level)